Hospital Nursing Intervention

Pilot Program

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ALY6030 Data Warehousing & SQL

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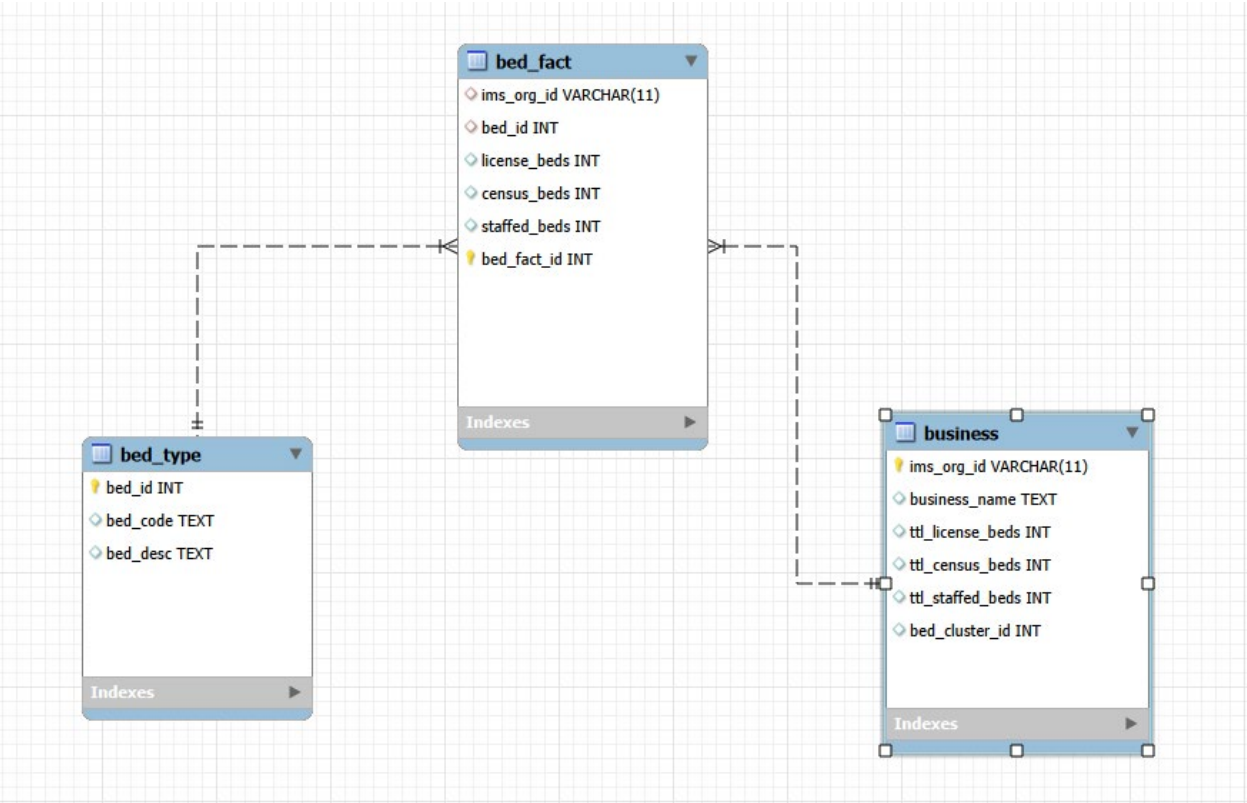
Step 1

For the bed_type table, all variables are dimensions because the table contains no numeric measures. The primary key bed_id identifies each bed type, while bed_code and bed_desc provide descriptive context. In the business table, the three dimensions are ims_org_id (the primary key), business_name, and bed_cluster_id, which categorizes hospitals. The remaining variables in the business table represent facts because they are numeric and aggregatable.

Step 2

In the bed_fact table, license_beds, census_beds, and staffed_beds are fact variables because they are numeric and aggregatable, while ims_org_id and bed_id function as dimensions since they are foreign keys that identify the hospital and bed type.

Step 3 – Star Schema



Step 4a: Analysis for Leadership (see SQL file for queries)

1. Licensed beds

hospital_name	total_license_beds
Phoenix Childrens Hospital	247
University of Maryland Medical Center	220
UC Health University Hospital	218
Wesley Medical Center, LLC	214
Vidant Medical Center	204
Rady Childrens Hospital and Health Center	200
Dallas County Hospital Association	195
Saint Lukes Episcopal Hospital Texas Medical Ce...	178
The Methodist Hospital	170
Emory University Hospital	169

2. Census beds

hospital_name	total_census_beds
Shands Hospital at the University of Florida	167
Dallas County Hospital Association	145
Mercy Medical Center Saint Louis	142
Los Angeles County University of Southern Calif...	139
The Methodist Hospital	138
University of Minnesota Medical Center Fairview	129
University of Maryland Medical Center	127
Brigham and Womens Hospital	124
Vidant Medical Center	123
Ronald Reagan University of California Los Ang...	122

3. Staffed beds

hospital_name	total_staffed_beds
Vidant Medical Center	203
Rady Childrens Hospital and Health Center	200
University of Maryland Medical Center	171
Emory University Hospital	169
Shands Hospital at the University of Florida	167
Mercy Medical Center Saint Louis	163
Wesley Medical Center, LLC	162
Phoenix Childrens Hospital	159
Grady Memorial Hospital	154
UC Health University Hospital	151

Step 4b: Interpretation of findings

In looking at the three summary tables above, only the University of Maryland Medical Center and Vidant Medical Center appear in all three lists. Phoenix Medical Center has the highest legal bed capacity with 247 licenses, and while they only have 117 beds on site (census_beds), they do have staffing for up to 159 beds (staffed_beds). Vidant Medical Center has the most staffed beds at 203, which is nearly all their 204 licenses but only a total of 123 current beds on site. To better explore which hospitals might be the best candidates for an additional staffing pilot, the percentage of staffing to census and licensed beds was calculated.

hospital_name	total_license_beds	total_census_beds	total_staffed_beds	pct_staffed_of_census	pct_staffed_of_license
Phoenix Childrens Hospital	247	117	159	135.90	64.37
University of Maryland Medical Center	220	127	171	134.65	77.73
UC Health University Hospital	218	110	151	137.27	69.27
Wesley Medical Center, LLC	214	98	162	165.31	75.70
Vidant Medical Center	204	123	203	165.04	99.51
Rady Childrens Hospital and Health Center	200	75	200	266.67	100.00
Dallas County Hospital Association	195	145	149	102.76	76.41
Saint Lukes Episcopal Hospital Texas Medical Ce...	178	99	124	125.25	69.66
The Methodist Hospital	170	138	138	100.00	81.18
Emory University Hospital	169	120	169	140.83	100.00
Shands Hospital at the University of Florida	167	167	167	100.00	100.00
Miami Valley Hospital	164	111	144	129.73	87.80
Mercy Medical Center Saint Louis	163	142	163	114.79	100.00
Los Angeles County University of Southern Calif...	160	139	151	108.63	94.38
MCGHealth, Inc	155	88	96	109.09	61.94

Here only The Methodist Hospital and Shands Hospital at the University of Florida did not have staffing levels in excess of census beds, while only 4 (Rady Children's, Emory University, Shands, and Mercy Medical Center Saint Louis) had staffing to accommodate all of their licensed beds.

Step 5a: Drill down investigation (see SQL file for queries)

1. Licensed beds

hospital_name	total_license_beds
University of Maryland Medical Center	220
UC Health University Hospital	218
Shands Hospital at the University of Florida	167
MCGHealth, Inc	155
Grady Memorial Hospital	154
Jackson Memorial Hospital	151
University Hospital in Bexar County	144
University of Minnesota Medical Center Fairview	144
Carolinas Medical Center	137
Yale New Haven Hospital	136

2. Census beds

hospital_name	total_census_beds
Shands Hospital at the University of Florida	167
University of Minnesota Medical Center Fairview	129
University of Maryland Medical Center	127
Jackson Memorial Hospital	117
UC Health University Hospital	110
Carolinas Medical Center	106
Cedars Sinai Health System	92
University Hospital in Bexar County	91
Duke University Health System	91
MCGHealth, Inc	88

3. Staffed beds

hospital_name	total_staffed_beds
University of Maryland Medical Center	171
Shands Hospital at the University of Florida	167
Grady Memorial Hospital	154
UC Health University Hospital	151
University of Minnesota Medical Center Fairview	144
Carolinas Medical Center	137
Saint Josephs Hospital and Medical Center	134
Chattanooga Hamilton County Hospital Authority	134
Jackson Memorial Hospital	128
Sunrise Hospital and Medical Center, LLC	125

Step 5b: Final recommendation

After recalculating the reports from step 4 to only include hospitals that have sufficient volume of both ICU and SICU beds, the staffing ratios were calculated for the top 15 hospitals by total licensed beds.

hospital_name	total_license_beds	total_census_beds	total_staffed_beds	pct_staffed_of_license	pct_staffed_of_census
University of Maryland Medical Center	220	127	171	77.73	134.65
UC Health University Hospital	218	110	151	69.27	137.27
Shands Hospital at the University of Florida	167	167	167	100.00	100.00
MCGHealth, Inc	155	88	96	61.94	109.09
Grady Memorial Hospital	154	88	154	100.00	175.00
Jackson Memorial Hospital	151	117	128	84.77	109.40
University Hospital in Bexar County	144	91	118	81.94	129.67
University of Minnesota Medical Center Fairview	144	129	144	100.00	111.63
Carolinas Medical Center	137	106	137	100.00	129.25
Yale New Haven Hospital	136	64	75	55.15	117.19
Saint Josephs Hospital and Medical Center	134	80	134	100.00	167.50
Chattanooga Hamilton County Hospital Authority	134	77	134	100.00	174.03
Sunrise Hospital and Medical Center, LLC	125	88	125	100.00	142.05
Hennepin Health Care System, Inc	125	47	69	55.20	146.81
Duke University Health System	113	91	97	85.84	106.59

When determining locations for an increased staffing pilot program, those that are at the lowest rates of staffed versus licensed beds would be ideal candidates. These represent locations with the capacity to add physical beds to their locations and staffing to provide care for patients who utilize those beds. From the report above, Yale New Haven Hospital and Hennepin Heath Care System, Inc stand out as being below 60% in the percentage staffed of licenses metric. As leadership has requested up to two pilot sites, those would be the recommended starting locations for the pilot program.